

Dante W. Mamani

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EDUCATION

Florida International University (FIU), Miami, Florida

Bachelor's Mechanical Engineering

April 2026

Relevant Coursework

- Statics (EGN 3311)
- Introduction to CAD for Mechanical Engineering (EML 1533)
- Thermodynamics (EGN 3343)
- Manufacturing Processes (EIN 3390)
- Mechanical Design I (EML 3500) and Mechanical Design II (EML 4501)
- Dynamics (EGN 3321)

Broward College, Davie, Florida

Associate of Arts Degree

April 2023

Relevant Course Work

- Differential Equations (MAP 2302)

Rey de Reyes, Arequipa, Peru

High school

December 2018

EXPERIENCE

Espumas Wflex Del Peru, Arequipa, Peru

Operations Support

May 2023 - July 2023

- Oversaw foam bed inventory flow by verifying quantities produced, shipped, and received, maintaining accurate production records.
- Facilitated basic sales tasks, including organizing products and preparing items for customer pickup.
- Provided friendly customer service during busy periods to help team meet demand.

Dicosur, Arequipa, Peru

Personal Assistant (Volunteer / Occasional)

April 2017 - March 2019

- Executed daily operational tasks for a store manager, including product organization, errands, and customer support.
- Maintained an orderly workspace and prepared items for customer needs.

PROJECTS EXPERIENCE

Water Distillation System, Miami, FL

Mechanical Engineering Student-Research & Analysis Engineer

September 2025 - April 2026

- Calculated heat-transfer requirements, material performance, and energy input needed for phase change.
- Estimated flow rate, production time, and system efficiency to support prototype design decisions.
- Created engineering drawings with dimensions, annotations, and material details operating SolidWorks.

Gear Pump Design - Mechanical Design II, Miami, FL

Mechanical Engineering Student (Design & Analysis)

August 2025 - November 2025

- Designed and modeled a functional gear pump operating SolidWorks.
- Conducted mechanical component analysis including stress distribution and system optimization.

Solar Powered Reverse Osmosis Desalination System - Design of Thermal and Fluid Systems, Miami, FL

Mechanical Engineering Student-Research & Analysis Engineer

August 2025 - November 2025

- Formulated a modular seawater reverse osmosis (RO) plant capable of producing 16.5 million gallons per day (MGD) of freshwater.

- Configured 2,368 RO membranes distributed across 16 systems with mixed-flow pump configurations.
- Developed system layout based on standard RO operating ranges and desalination literature.
- Integrated solar energy concepts into desalination design for sustainable freshwater production.
- Prepared conceptual cost estimation using standardized bid structure.
- Evaluated large-scale water resource challenges and engineering feasibility.

TECHNICAL SKILLS

- Engineering Sketching
- Technical Drawing
- Microsoft (Excel, PowerPoint, Word)
- CAD modeling (SolidWorks, and Onshape)
- Python (simulations and calculations)
- Matlab (calculations)

LANGUAGES

- Spanish (verbal and written)
- English (Verbal and written)